

TACNA, Peru

WMO: 847820

Lat: 18.053S

Lon: 70.276W

Elev: 1538

StdP: 13.90

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	49.7	50.3	46.1	48.7	52.4	46.7	49.9	52.7	14.2	63.7	13.1	63.3	2.6	30	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.1	83.9	69.1	82.3	68.1	80.7	67.2	70.1	81.3	68.8	79.7	67.9	78.6	11.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	101.9	76.2	64.6	96.7	75.2	63.7	93.4	74.6	34.9	81.6	33.9	80.3	33.1	78.9	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	14.0	12.7	DB	47.3	86.8	2.2	1.6	45.8	87.9	44.5	88.9	43.3	89.8	41.7	91.0	(4)
(5)			WB	45.6	71.4	2.2	2.1	44.0	72.9	42.7	74.1	41.4	75.2	39.8	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.0	72.7	74.0	72.1	68.4	64.3	60.9	59.1	59.5	61.1	63.8	66.6	69.5
	DBStd	5.56	2.11	2.13	2.13	2.36	2.64	2.30	2.20	2.61	2.42	2.10	1.76	1.76
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	702	0	0	0	3	44	123	184	174	118	50	6	0
	CDD50	5827	705	673	686	551	444	328	281	293	334	427	499	606
	CDD65	1053	240	253	221	104	23	1	0	2	2	11	55	141
	CDH74	4008	1037	1240	899	266	38	4	2	11	3	19	95	394
	CDH80	457	128	210	94	8	2	1	1	2	0	1	0	10
Wind	WSAvg	5.0	5.5	5.6	5.2	4.7	4.3	4.2	4.3	4.5	4.9	5.6	5.8	5.6
Precipitation	PrecAvg	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0
	PrecMax	1.6	0.6	0.2	0.1	0.2	0.2	0.2	0.4	0.6	0.6	0.4	0.1	0.3
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.2	0.1	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.4	86.2	84.3	80.9	77.3	73.0	70.1	73.3	73.7	76.2	78.5	81.0
		MCWB	70.4	70.5	70.0	67.2	65.0	61.9	60.0	62.8	62.2	63.5	64.5	66.9
	2%	DB	82.8	84.2	82.4	78.8	74.8	70.0	67.9	69.4	71.5	73.6	76.6	79.1
		MCWB	68.3	69.2	68.6	66.2	64.0	60.5	59.3	59.6	60.7	62.0	63.5	65.8
	5%	DB	81.0	82.5	80.7	77.0	72.0	68.1	66.0	67.2	69.4	71.8	75.1	77.4
		MCWB	67.3	68.3	67.5	65.0	62.2	59.8	58.0	58.6	59.7	61.0	63.0	64.7
	10%	DB	79.3	80.8	79.0	75.2	70.1	66.2	64.2	64.8	67.2	70.0	73.4	76.7
		MCWB	66.3	67.4	66.6	64.5	61.3	58.6	57.0	57.2	58.8	59.9	62.3	64.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.5	71.7	70.9	68.2	65.7	62.3	61.2	63.1	62.9	63.9	65.1	68.1
		MCDB	82.9	84.2	82.5	78.7	75.8	71.1	68.5	71.9	71.5	74.6	76.6	78.9
	2%	WB	69.7	70.2	69.4	66.8	64.0	60.9	59.4	60.3	61.4	62.6	64.0	66.6
		MCDB	80.2	81.9	80.7	77.4	73.4	69.0	66.0	67.1	69.4	72.4	75.1	77.7
	5%	WB	68.2	69.0	68.1	65.5	62.9	59.8	58.4	58.9	60.1	61.4	63.0	65.6
		MCDB	78.7	80.2	78.9	76.0	71.3	66.9	64.3	66.0	67.6	70.8	73.7	76.4
	10%	WB	67.2	68.0	67.1	64.4	61.7	58.8	57.4	57.6	59.0	60.3	62.2	64.6
		MCDB	77.8	78.8	77.6	74.6	69.1	64.9	62.7	64.2	66.3	69.1	72.2	74.9
Mean Daily Temperature Range	5% DB	MDBR	16.9	17.1	17.3	16.5	14.9	13.4	13.3	14.3	15.0	16.1	16.5	16.9
		MCDBR	17.6	17.6	17.8	17.7	16.3	15.1	14.8	16.0	17.1	17.2	17.5	17.4
		MCWBR	8.0	7.6	8.2	8.5	8.0	7.6	7.6	8.0	8.3	7.9	7.9	7.9
	5% WB	MCDBR	16.8	16.9	17.3	17.1	15.9	14.5	14.1	15.6	16.3	16.7	16.9	16.9
	MCWBR	7.9	7.4	8.1	8.3	7.9	7.5	7.5	8.0	8.1	7.8	7.7	7.8	
Clear-Sky Solar Irradiance	taub	0.504	0.507	0.491	0.435	0.373	0.354	0.354	0.379	0.401	0.426	0.428	0.477	
	taud	2.053	2.052	2.093	2.208	2.326	2.385	2.376	2.279	2.217	2.150	2.175	2.080	
	Ebn at Noon	270	266	263	266	272	272	276	279	284	285	289	277	
	Edh at Noon	57	57	53	44	36	33	34	40	46	51	50	56	
All-Sky Solar Radiation	RadAvg	2298	2174	2141	1887	1599	1376	1393	1651	1959	2228	2481	2460	
	RadStd	167	175	131	127	74	79	86	90	106	86	100	116	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page